Lead-Time Analysis and Logistics Optimization Report: ABC Company

University Final Project
Introduction to Programming and Data Management in R
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Executive Summary

Amidst the COVID-19 pandemic, ABC company experienced extended manufacturing and shipping lead times for the products risking significant customer dissatisfaction. ABC Company initially requested an analysis of the relationship between different manufacturers and ship modes on logistics lead times to improve predictability in the logistics and transportation of their products.

To aid in the understanding of extended lead times, this data analysis report aimed to answer several vital questions:

- Which original design manufacturer (ODM) site has the highest manufacturing delay?
- Which ODM site has the highest number of outliers for manufacturing and in-transit lead times?
- Do certain quarters show higher in-transit lead times and unpredictability?
- How does the "In-transit Lead Time" vary by ship mode and which ship mode is the fastest?
- Does the line of business (LOB) affect manufacturing time?

Upon analyzing the findings, several recommendations were determined:

- **Optimize Allocation:** Prioritize time-sensitive orders through Sites C and D. These sites remained the most consistent and had the lowest manufacturing and in-transit times even during the Q3/Q4 peak delay periods.
- **Ship Mode Selection:** Shift away from water-based logistics where possible. Ground and Air should be the preferred modes, as ocean and fast boat transit poses a high risk for significant delays.
- **Manage Expectations for Product C:** Given that Product C requires more time to manufacture than Products A and B, expect higher lead-time estimates specifically for this product.
- **Expect Holiday Delays:** Set customer expectations going forward to expect manufacturing and/or shipment delays during Q4.
- **Buffer Site A:** Site A demonstrates the widest spread of manufacturing and in-transit lead times in the network. It is recommended to widen the promised delivery to account for the variability.

1. Project Overview

ABC, a major technology company that produces desktops and laptops, experienced extreme instability within the supply chain of its products amidst the COVID-19 pandemic. This volatility resulted in delayed manufacturing and shipping lead times, which risked significant customer dissatisfaction. To improve the customer experience, ABC company requested an analysis of the relationship between different manufacturers and ship modes on logistics lead times.

2. Data Overview

2.1 Data Preparation & Cleaning

ABC company provided a dataset comprising six variables with information across three lines of business (Product A, B, C), four factories (Site A, B, C, D), and four ship modes (ground, air, ocean, fast boat). The raw data also included purchase order (PO) download dates, ship dates, and receipt dates. Additionally, I utilized a provided fiscal calendar detailing the date ranges for quarter and year information.

The raw data set began with 9,124 observations across six columns. To prepare the data for analysis, I calculated the “Manufacturing Lead Time” as the difference in days between the PO download and ship dates, and I calculated the “In-transit Lead Time” as the difference in days between the receipt and ship dates. My cleaning process addressed two primary problematic values: negative lead times caused by out-of-order date entries and missingness across the logistics information.

After I calculated the initial manufacturing and transit lead times, I identified a total of 331 rows with negative lead time values. This indicated that the download, ship, and receipt dates were not in ascending order. To correct this, I developed a custom R function named “sort_dates_chronologically” that united the date columns, sorted the strings in ascending order, and separated them back into their corresponding columns. Upon my recalculation of the lead times, the error was resolved and only positive values remained for manufacturing and delivery time calculations.

Next, I integrated the fiscal calendar by attaching quarter and year information based on the receipt date. After this step, the dataset contained around 1008 missing values spread across 6 columns: Ship Date, Receipt Date, In-transit Lead Time, Manufacturing Lead Time, Quarter, and Year. To remediate this, I imputed the median in-transit and manufacturing lead times by grouping the data by origin, line of business (LOB), and ship mode. I selected the median

because it resisted skewing from outliers more effectively than the mean. Using these medians, I calculated the missing ship date values by adding the Manufacturing Lead Time to the PO Download Date, and I derived missing receipt dates by adding the In-transit Lead Time to the Ship Date.

These imputations left 183 missing values in each of the Quarter and Year columns. I resolved these missing values by cross-referencing the fiscal calendar and repeating the previous join approach specifically for those gaps. Using a date-range join, I identified the correct quarter and year values to fill the remaining gaps without overwriting existing data. Following this step, only 10 rows remained with missing values in the Quarter and Year columns. Because these observations lacked a corresponding date range in the fiscal calendar, I removed them. I assessed that this action posed little risk to the statistical analysis.

After I finished the data cleaning process, the final dataset contained 9,114 observations across seven variables: LOB, Origin, Ship Mode, In-transit Lead Time, Manufacturing Lead Time, Quarter, and Year. In total, I removed 10 rows and I added two columns, “Quarter” and “Year,” to facilitate the analysis.

2.2 Statistical Measurements

To measure central tendency, I chose the median, the middle value of the data, over the mean because it offered a better representation for a dataset containing outliers.

To measure the spread or statistical variability of the middle half of the data, I calculated the Interquartile Range (IQR). The IQR represented the distance between the 75th and 25th percentiles and proved useful for identifying the typical range of datapoints without the impact of outliers. In this analysis, I used the IQR to measure the consistency of routine manufacturing and delivery times by representing the typical performance range for each site.

Alongside the IQR, I calculated the proportion of outliers relative to total shipments across site origins to measure volatility. These percentages quantified how often extreme delays occurred. I prioritized column charts over box plots for outlier visualization. Due to the extreme range of lead times and significant overplotting in the distribution's long tail, I used a percentage-based approach to more effectively quantify and compare the frequency of extreme delays across sites.

To measure the correlation between in-transit lead time and categorical values, I used a process called one-hot encoding. During this process, I converted the Origin, LOB, Ship Mode, and Quarter columns into numeric values that I compared against in-transit lead times.

3. Findings

Table 1. Manufacturing Lead Time Summary

Minimum	1st Quarter	Median	Mean	3rd Quarter	Maximum
0.0	4.0	6.0	8.2	11.0	143.0

Table 2. In-Transit Lead Time Summary

Minimum	1st Quarter	Median	Mean	3rd Quarter	Maximum
0.0	5.0	8.0	14.8	23.0	235.0

3.1 Central Tendency Analysis

The goal of this section is to provide insight into which Original Design Manufacturers (ODM) Site has the highest manufacturing and in-transit delays. To analyze how the manufacturing and in-transit lead time vary by site origin, the analysis aggregates lead-time data by ODM site to establish a baseline for both manufacturing and in-transit performance. Tables were arranged in ascending order of their mean lead time.

Figure 1. Median Manufacturing Lead Time by Site Origin

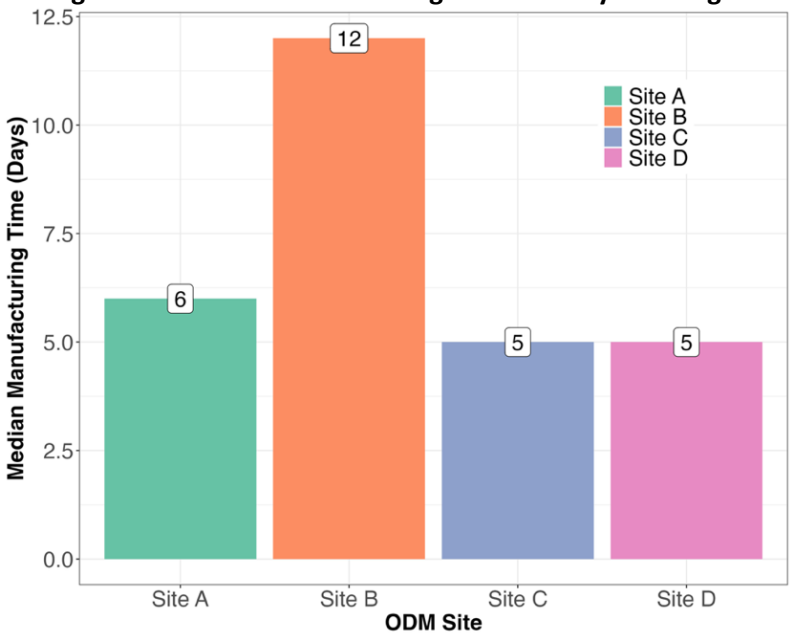


Table 3. Mean and Median Manufacturing Lead Time by Site Origin

Origin	Mean (Days)	Median (Days)	Observations
Site C	5.8	5	1650
Site D	6.5	5	2432
Site A	8.4	6	3464
Site B	12.9	12	1568

Note. Red cells indicate the highest lead times, while green cells indicate the lowest lead times.

Analysis:

- **Site C** maintains the shortest median manufacturing lead time of 6 days and a mean of 5.8 days compared to all sites. This makes Site C a top choice for fast manufacturing times.
- **Site D and A** maintain median lead times within reason of 5 and 6 days. This places them as the next best alternative sites for efficient manufacturing speed behind Site C.
- **Site B** generates the highest median and mean manufacturing lead time of 12 and 12.9 days. These elevated metrics identify Site B as a primary source of manufacturing delays.

Figure 2. Median In-Transit Lead Time by Site Origin

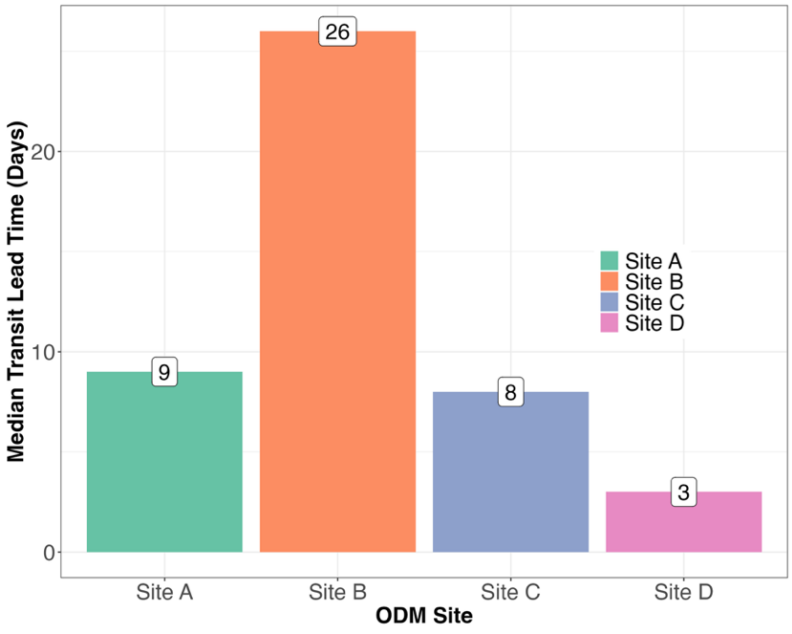


Table 4. Mean and Median In-Transit Lead Time by Site Origin

Origin	Mean (Days)	Median (Days)	Observations
Site D	4.6	3	2432
Site C	9.7	8	1650
Site A	19.2	9	3464
Site B	26.3	26	1568

Analysis:

- **Site D** has the lowest mean and median in-transit lead times. This makes it the strongest choice for prioritizing efficient delivery speed.
- **Site C** maintains relatively low in-transit lead times compared to the delivery durations of site A and B. While its in-transit times slightly exceed those of Site D, Site C remains a highly reliable option for rapid shipment.
- **Site A** exhibits a right-skewed distribution where the mean in-transit lead time of 19.2 more than doubles the median of 9. This discrepancy confirms the presence of significant outliers and suggests a risk for delivery reliability.
- **Site B** generates the highest in-transit lead times. These elevated metrics identify Site B as the primary source of operational delays, with its median delivery time taking 26 days. This designates it as the riskiest site origin for any products requiring expedited delivery.

3.2 Variability Analysis

To assess the reliability of site operations, this section examines the Interquartile Range (IQR) of both manufacturing and in-transit lead times. While the central tendency (mean and median) describes the average time, the IQR measures the spread of the middle 50% of the data. In the context of this report, a smaller IQR indicates a more consistent and predictable process, and a larger IQR suggests more fluctuations in the expected lead times.

Table 5. Interquartile Range of Manufacturing Lead Time by Site Origin

Origin	IQR	Observations
Site D	3	2432
Site B	4	1568
Site C	4	1650
Site A	6	3464

Note. Red cells indicate the IQR high, while green cells indicate the IQR low.

Table 6. Interquartile Range of In-Transit Lead Time by Site Origin

Origin	IQR	Observations
Site D	2	2432
Site C	2	1650
Site B	5	1568
Site A	20	3464

Analysis:

- **Site A** demonstrates the most volatile manufacturing and in-transit lead times across the entire network. Its elevated IQR across both manufacturing and shipment categories reveals a significantly wider gap between the 25th and 75th percentiles compared to other locations. This spread suggests that Site A struggles with inconsistent processes and outlier values that lead to unpredictable delays.
- **Sites B, C, and D** achieve much tighter and more predictable lead times than Site A. Because the middle 50% of their data points cluster closely together, these sites likely maintain more stable manufacturing processes and shipment fulfillment. These findings identify Site D as the most reliable option for meeting delivery and shipment expectations, followed by sites B and C.

3.3 Outlier Analysis

Identifying the proportion of outlier manufacturing and delivery times by site origin is important in determining the level of risk a site poses. Although a site origin may have a low IQR with the bulk of its data points relatively close to one another, the percentage of outliers outside this range can provide us insight into what happens when unexpected delays occur.

To calculate outliers, I first determined the 25th and 75th percentile. Then I calculated the upper and lower bounds using the following equations:

- Lower Bound = Q1 – (1.5 x IQR)
- Upper Bound = Q3 + (1.5 x IQR)

Any value lower than the lower bound or higher than the upper bound was counted as an outlier.

Figure 3. Percent of Manufacturing Lead Time Outliers by Site Origin

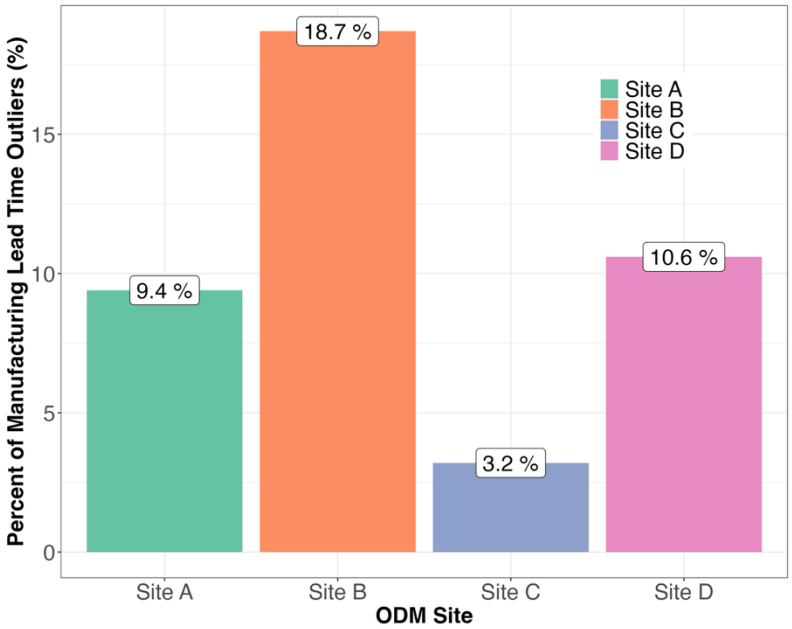


Figure 4. Percent of In-transit Lead Time Outliers by Site Origin



Analysis:

- **Site C** maintains the lowest percentage of manufacturing outliers (3.2%), making it the most stable origin for production. Conversely, site C has the highest proportion of in-transit lead time outliers (5.3%). However, its relatively small proportion of outliers indicates that it poses a low risk for delivery unpredictability.
- **Site A** is the most reliable for shipping, with the lowest percentage of in-transit outliers (1.9%).

- **Site B** represents the highest operational risk for manufacturing, with nearly one-fifth of all orders (18.7%) resulting in outlier lead times.

3.4 Temporal Trends

This analysis established whether certain quarters show higher delays than others and measured the amount of product deliveries. To analyze the relationship between quarter and lead times (figures 5-8) and calculate the number of orders delivered (figure 9), the quarters from the fiscal calendar were aligned to receipt dates. This allowed for a chronological view of performance shifts.

Figure 5. Interquartile Range of Manufacturing Lead Times by Quarter

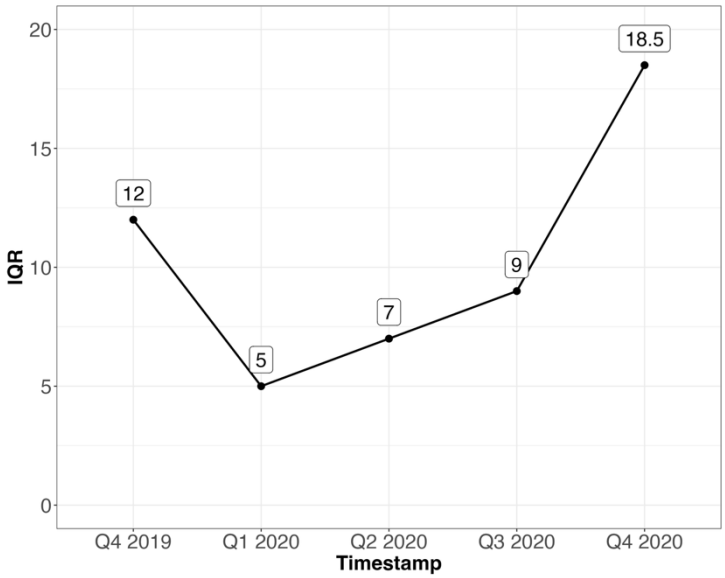


Figure 6. Interquartile Range of In-transit Lead Times by Quarter

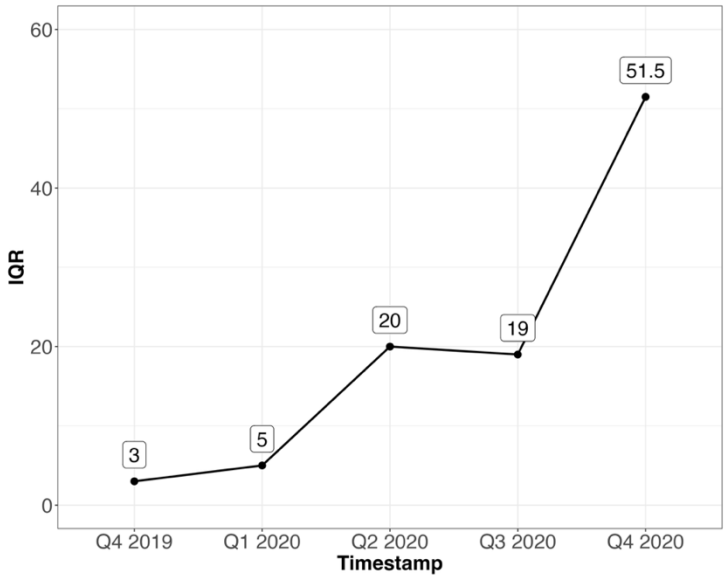


Figure 7. Median Manufacturing Lead Time by Quarter and Site

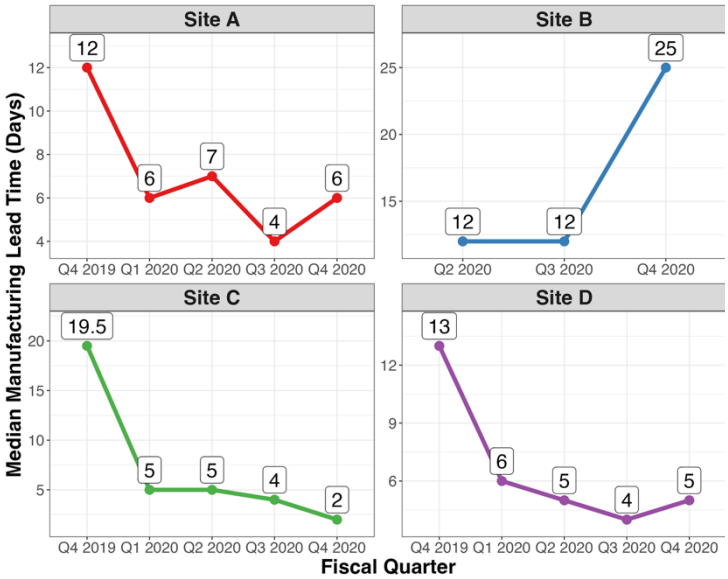
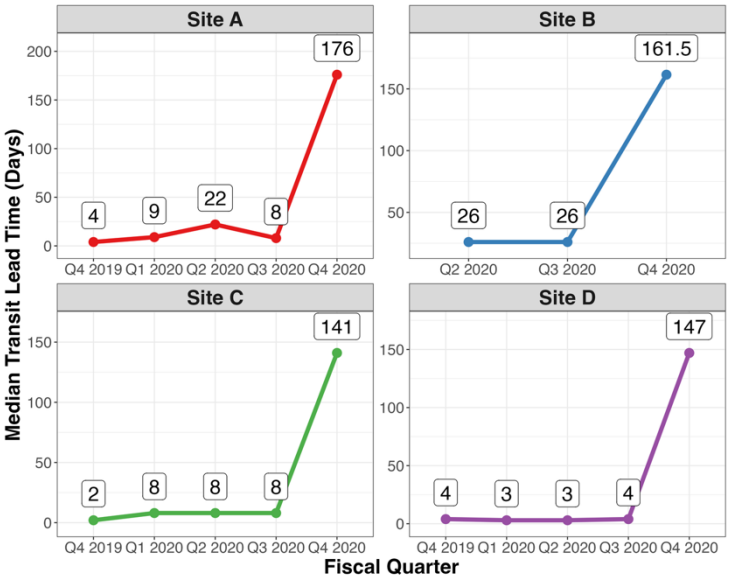


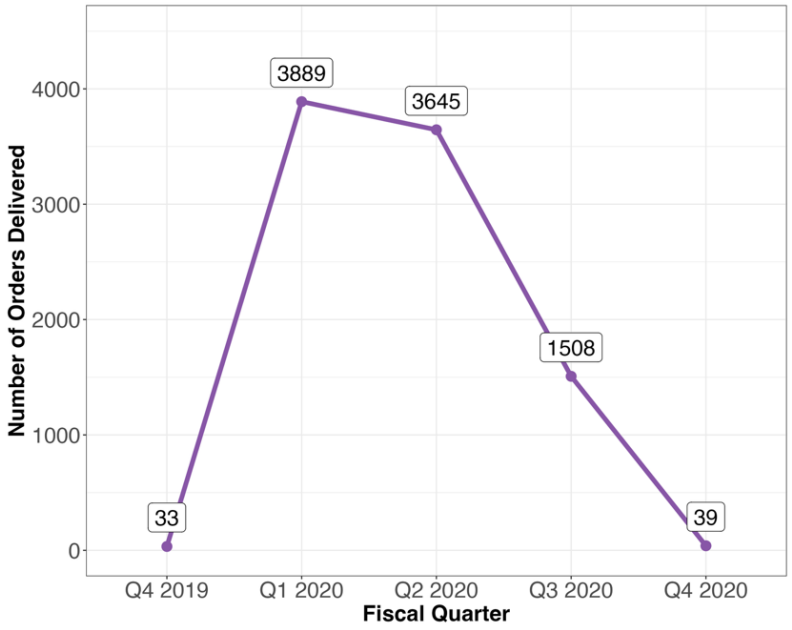
Figure 8. Median In-Transit Lead Time by Quarter and Site



Analysis:

- **The expansion of the IQR** for both in-transit and manufacturing lead times between Q3 and Q4 2020 signals declining process consistency and suggests a higher frequency of statistical outliers during the final quarter.
- **A significant spike in median transit times** across all site origins in Q4 2020 indicates a systemic slowdown in distribution networks.
- **Median manufacturing trends between sites** show that Sites A and B drove the upward lead-time trend during Q4 2020, whereas all other sites either maintained stability or achieved slight reductions.
- **The data identifies Q4 2020** as the peak period for both manufacturing and in-transit delays across the entire organizational network.

Figure 9. Customer Orders Delivered per Quarter



Analysis

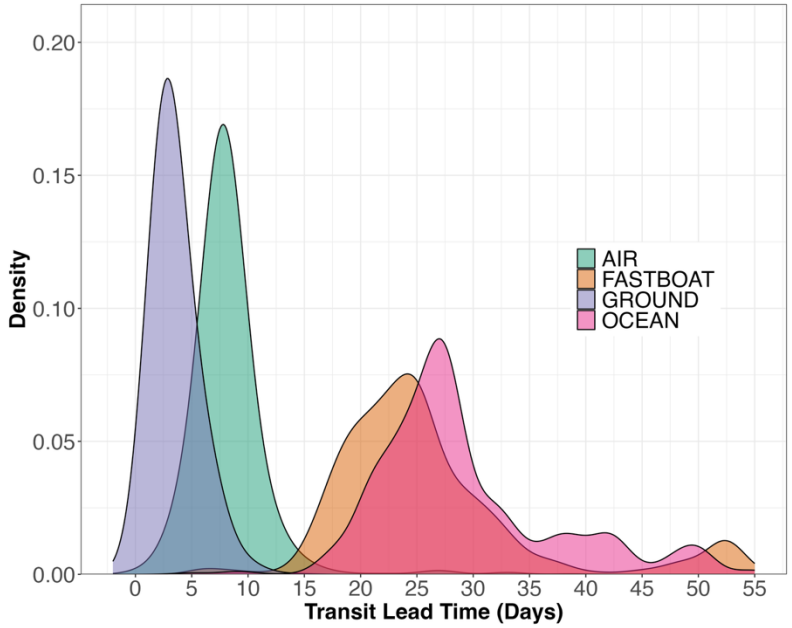
- **Figure 9 showcases a collapse in delivered orders.** The number of orders delivered saw volumes plummet from 1,508 in Q3 2020 to just 39 in Q4 2020. This estimated 97% decline in order receipts may result from a combination of suppressed customer demands between Q3 and Q4 of 2020 or manufacturing delays indicated earlier.

3.5 Categorical Relationships

3.5.1 Ship Mode Effects

To identify which Ship Mode has the longest median delivery lead times and most unpredictability, or largest IQR, the analysis isolated in-transit performance by Ship Mode to identify if specific ship modes were more efficient than others. Tables were arranged in ascending order by their median lead time and IQR to highlight the efficiency of various ship modes.

Figure 10. Distribution of In-Transit Lead Time by Ship Mode



Note. The x-axis is truncated at 55 days to highlight the primary data density and improve readability.

Table 7. Median and IQR of In-Transit Lead Time by Ship Mode

Ship Mode	Median Lead Time (Days)	IQR	Count
Ground	3	2	2432
Air	8	2	3810
Fast boat	24	7.8	530
Ocean	27	9	2342

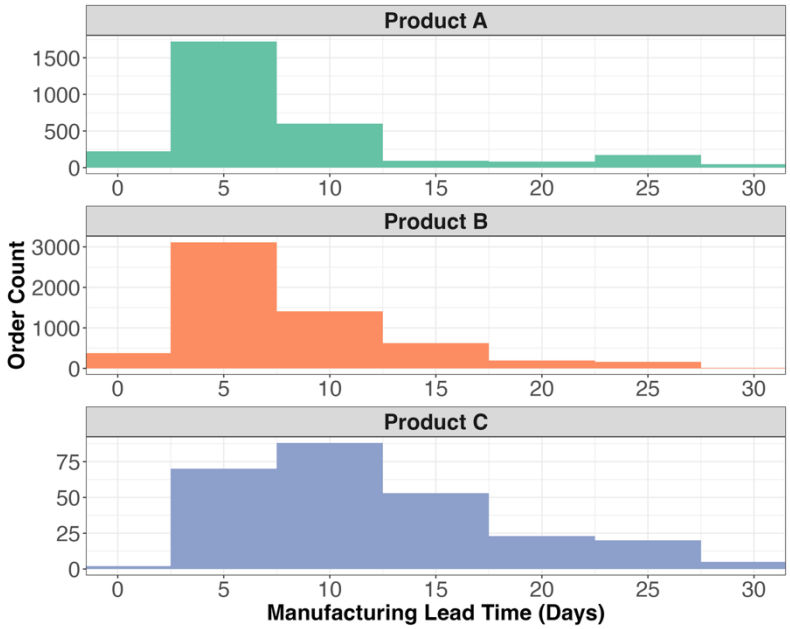
Analysis:

- **Ocean and Fast boat** exhibit the widest and flattest density curves and thus signal high logistics unpredictability. These ship modes are more spread out and across a 15-to-50-day window, so this signals they carry the highest risk for long-term delays. Additionally, these two ship modes possess the highest median transit delivery times with Fast boat at 24 days and Ocean at 27 days.
- **Ground** provides the most reliable transportation service. Its narrow and tall peak indicates that most shipments arrive very close to the median lead time. Ground maintains the lowest IQR of 2 and the lowest median of 3 days. In summary, ground drastically outperforms the "flat" distributions of other ship modes that spread over multiple weeks.
- **Air** follows Ground in being highly reliable and demonstrates consistent delivery performance through its tight density curve.

3.5.2 Line of Business (LOB) Effects

To analyze which Products had the highest manufacturing delays, this analysis isolated lead times by Line of Business (LOB) to identify if specific products received higher assembly priority. Tables were arranged by their median performance and IQR, highlighting which product lines moved most efficiently through the manufacturing phase.

Figure 11. Distribution of Manufacturing Lead Time by Line of Business (LOB)



Note. The x-axis is truncated at 30 days to highlight the primary data density and improve readability.

Table 8. Manufacturing Lead Time by Line of Business (LOB)

LOB	Median Lead Time (Days)	IQR	Count
Product A	6	5	2944
Product B	6	7	5908
Product C	10	9	262

Analysis

- **Products A and B** prioritize manufacturing speed by achieving the lowest median lead time of 6 days. Product A demonstrates superior process stability by maintaining the lowest IQR of 5. This tighter data cluster indicates a more consistent production cycle compared to Product B.
- **Product C** requires the longest manufacturing duration and produces the longest median lead time of 10 days. This extended timeline suggests either a lower production priority or a more complex assembly process. Furthermore, Product C generates the highest IQR (9) and is thus identified it as the most unpredictable product regarding manufacturing lead times.

3.6 Correlation Analysis

To identify the relationships between various operational factors and delivery speed, a correlation analysis was performed on all encoded variables. By converting categorical data such as Line of Business, Origin, and Ship Mode into numeric format, this analysis could mathematically measure the strength and direction of each factor's impact on In-transit Lead Time. The resulting correlation table provides an overview of how these variables interact, and highlights which specific drivers are most strongly associated with increased delays or improved transit efficiency.

In the context of this dataset, these are the scales used for identifying the level of correlation:

- Strong Correlation |0.50| to |1.0|
- Moderate Correlation |0.30| to |0.49|
- Weak Correlation |0.10| to |0.29|
- Negligible/No Correlation 0.00 to |0.09|

Table 9. Correlation of Predictors with In-transit Lead Time

Variable	Correlation	Impact on Lead Time
Ship Mode: OCEAN	0.5133	Strongest Positive Impact (Longest Delays)
Quarter: Q4	0.3500	Moderate Positive Impact
Origin: Site B	0.2848	Moderate Positive Impact
Origin: Site A	0.1863	Weak Positive Impact
Ship Mode: FASTBOAT	0.1666	Weak Positive Impact
Quarter: Q3	0.1586	Weak Positive Impact
LOB: Product C	0.0918	Negligible Impact
Quarter: Q2	0.0626	Negligible Impact
LOB: Product A	0.0443	Negligible Impact
LOB: Product B	-0.0755	Negligible Impact
Origin: Site C	-0.1305	Weak Negative Impact
Ship Mode: AIR	-0.2345	Moderate Negative Impact (Faster)
Quarter: Q1	-0.2438	Moderate Negative Impact (Faster)
Origin: Site D	-0.3338	Strong Negative Impact (Fastest)
Ship Mode: GROUND	-0.3338	Strong Negative Impact (Fastest)

Note. Correlation numbers were rounded to 4 decimal places.

Analysis:

- Ocean ship mode has the highest positive correlation (0.51) and is thus the biggest delayer. This means when things go by ocean, the lead time increases significantly.
- Ground ship mode and ODM Site D have the strongest negative correlations (-0.33). This means these factors are associated with shorter (faster) lead times.
- Products A, B, and C has almost zero correlation, which means the line of business little association with in-transit lead times.

4. Conclusion

ABC Company initially requested an analysis of the relationship between different manufacturers and ship modes on logistics lead times to improve predictability in the logistics and transportation of their products.

To aid in the understanding of extended manufacturing and ship lead times and improve customer satisfaction, this analysis aimed to answer several vital questions:

- Which original design manufacturer (ODM) site has the highest manufacturing delay?
- Which ODM site has the highest number of outliers for manufacturing and in-transit lead times?
- Do certain quarters show higher in-transit lead times and unpredictability?
- How does the "In-transit Lead Time" vary by ship mode and which ship mode is the fastest?
- Does the line of business (LOB) affect manufacturing time?

Several findings were concluded:

- Site C led the network in manufacturing speed, maintaining the lowest average of 5.8 days. Site D operated as the most efficient origin for expedited delivery. Its median in-transit lead time ranked lowest in the network as many customers received orders only 3 days after shipment.
- Site B recorded the highest median manufacturing lead time (12 days) and in-transit lead time (26 days). These numbers designated the site as a significant risk to fast product handling.
- In terms of reliability, Site D offered the most consistent option for meeting lead time expectations. It demonstrated the lowest Interquartile Range (IQR) across both manufacturing and delivery categories. Additionally, Site C produced the fewest manufacturing outliers, while Site A generated the fewest in-transit outliers.
- The data identified Q4 2020 as the peak period for delays across the entire network. Notably, delivery volumes plummeted from 1,508 in Q3 2020 to just 39 in Q4 2020. This suggested a significant disruption or shift in operations during that period.
- Regarding transit methods, ground and air provided the quickest in-transit delivery times. Ocean remained the most significant bottleneck with a median in-transit time of 27 days, while fast boat followed closely behind.
- Finally, Product C represented a disproportionate manufacturing burden. It required a median production time of 10 days, which exceeded the median production times of both Product A and Product B by 4 days.

Recommendations:

1. **Optimize Allocation:** Prioritize time-sensitive orders through Sites C and D. These sites remained the most consistent and had the lowest manufacturing and in-transit times even during the Q3/Q4 peak delay periods.
2. **Ship Mode Selection:** Shift away from water-based logistics where possible. Ground and Air should be the preferred modes, as ocean and fast boat transit poses a high risk for significant delays.
3. **Manage Expectations for Product C:** Given that Product C requires more time to manufacture than Products A and B, expect higher lead-time estimates specifically for this product.
4. **Expect Holiday Delays:** Set customer expectations going forward to expect manufacturing and/or shipment delays during Q4.
5. **Buffer Site A:** Site A demonstrates the widest spread of manufacturing and in-transit lead times in the network. It is recommended to widen the promised delivery to account for the variability.